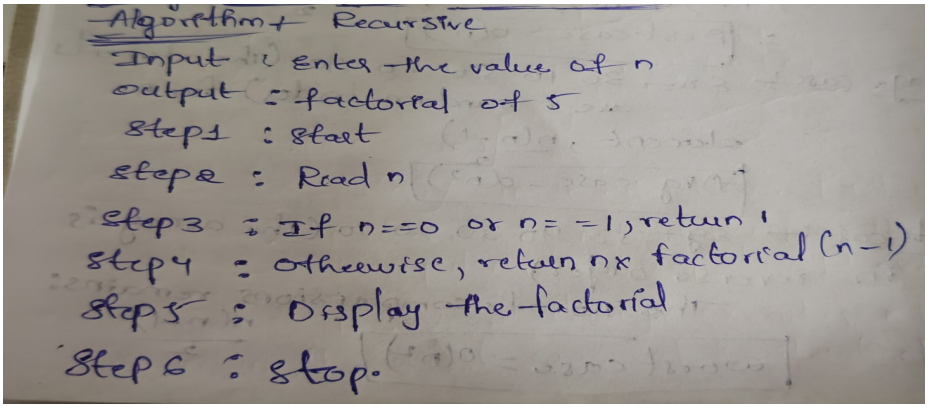
Practical-4

Aim: Implementation and Time analysis of factorial program using iterative and recursive method.

Algorithm:



Program:

1)Recursive method:

```
#include <iostream>

using namespace std;


int factorial(int n)

{

   if (n == 0 || n == 1)

      return 1;

   else

      return n * factorial(n - 1);

}


int main()
```

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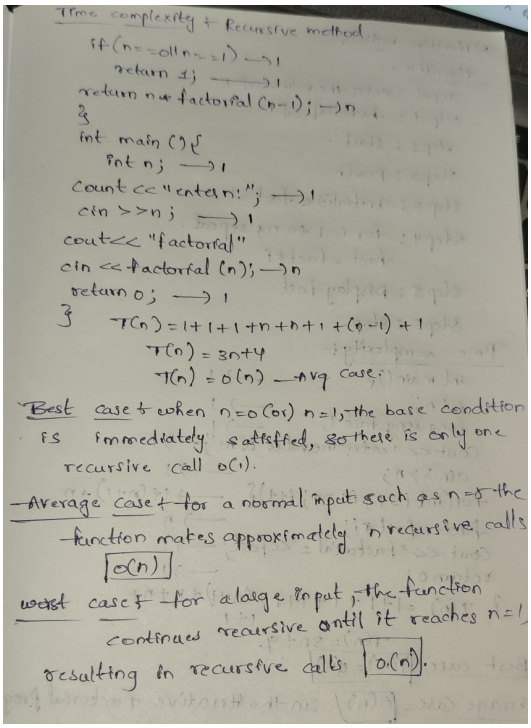
```
{
   int n;

   cout << "Enter a number: ";
   cin >> n;

   cout << "Factorial using Recursive Method = " << factorial(n);
return 0;
}
```

Output:

```
Enter a number: 5
Factorial using Recursive Method = 120

=== Code Execution Successful ===
```

Time complexity:



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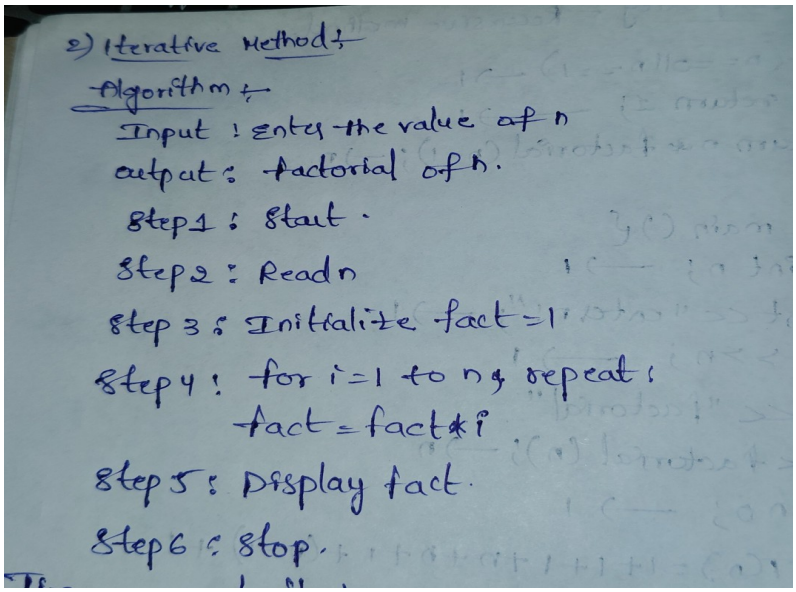
2) Iterative Method:

Algorithm:



Program:

```
#include <iostream>

using namespace std;


int main()

{

   int n, fact = 1;


   cout << "Enter a number: ";

   cin >> n;


   for (int i = 1; i <= n; i++)

   {

      fact = fact * i;

   }
```

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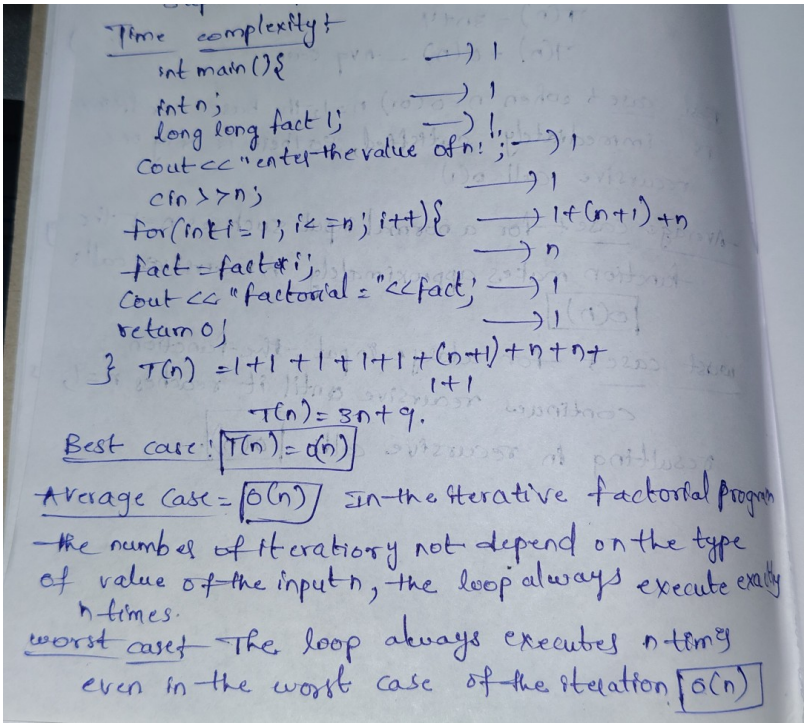
```
    cout << "Factorial using Iterative Method = " << fact;


    return 0;
}
```

Output:

```
Enter a number: 3
Factorial using Iterative Method = 6

=== Code Execution Successful ===
```

Time complexity:



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Conclusion:

The factorial program was successfully implemented using both iterative and recursive methods. The iterative method has O(n) time and 0(1) space complexity, while the recursive method has O(n) time and O(n) space complexity due to the recursive call stack.

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